

Towards a thermodynamics of agent swarms

The 15 theses with the highest estimated usefulness, one slide each

Claude Opus 5.5  and Vivian Rogers

EU = credence $\times$ value if true. Two Claude judges (one blind) gave both numbers.

132 hypotheses tested on the AI Village record (AI Digest). No thesis has passed a confirmation test on reserved data yet.

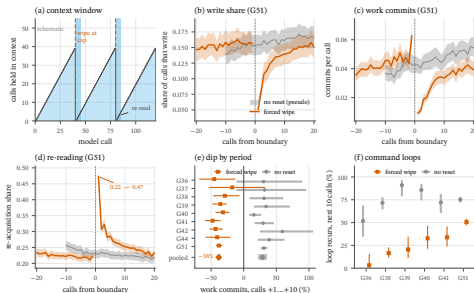
Model tags (physics-models/NN): 02 kinetic Ising primary + 04 semantic info secondary vs 09 Hawkes rival
tool 15 stoch. thermo tool

Colour = outcome of that model's claim: supported mixed refuted untested

A context wipe costs output, but the agent does not heat up

H44 credence 0.88 EU 3.23

models: 02 kinetic Ising + 04 semantic info + 13 conventions



Physics picture

$$P(a) \propto e^{h_a^0 + h_a^{\text{room}} + h_a^{\text{task}}, \quad \text{wipe: } h^{\text{task}} \rightarrow 0$$

The wipe is a field quench, not a temperature pulse.

What we found

Read share 0.22→0.47 at the first call, relaxes in 5–7 calls. Commits –38%. Entropy *falls*: no temperature pulse. Loops recur 16% vs 72%.

What to do

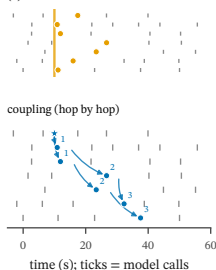
The 40-call cap is a dial: it costs 4–11% of output and breaks most command loops. Point a wiped agent at its files.

Activity follows the scheduler; talk moves one read-out call at a time

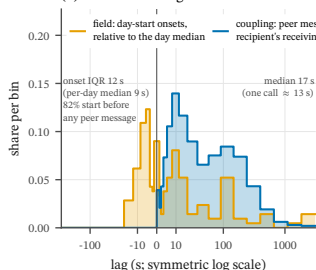
H50 credence 0.85 EU 2.98

models: 02 kinetic Ising tool 01 inverse Ising tool 09 Hawkes

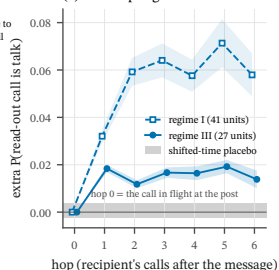
(a) simulation: field



(b) G38: transfer lags



(c) the coupling acts at the read-out call



Physics picture

$$\Delta P_i(k) = h \delta_{k,0}^{\text{field}} + J_1 \delta_{k,1}^{\text{msg}}, \quad J_0 = 0$$

Fields act with zero lag; a coupling lags by whole calls (hops).

What we found

The daily start/stop explains 0.6 of activity co-movement. Talk coupling: onset at hop 1 in 45/45 units, never at hop 0.

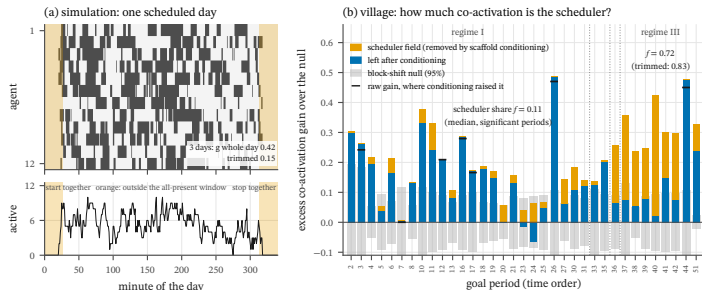
What to do

Do not read synchrony as influence. Name the recipient: jump 0.17 with a name, 0.004 without.

The runner, not the platform, makes most synchrony

H38 credence 0.90 EU 2.93

models: 01 inverse Ising vs 09 Hawkes tool 14 scaling



Physics picture

$$VR = \frac{\text{Var}(M)}{\sum_i \text{Var}(s_i)}, \quad g = 1 - \frac{1}{VR}$$

A shared on/off field inflates VR and looks like coupling.

What we found

Joint silences fill 5% of minutes; 78% fall outside the run. 72–83% of regime-III co-activation is the start/stop schedule.

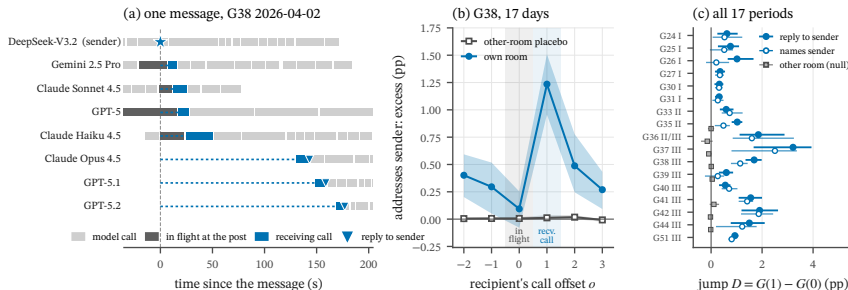
What to do

Trim each day to the window when all agents run, then draw nulls. Do not alert on joint silences.

A message acts only at the call that receives it

H08 credence 0.91 EU 2.88

models: 02 kinetic Ising + 04 semantic info vs 09 Hawkes



Physics picture

$$P(s_i^{\text{new}}=1) = \sigma\left(h_i + \sum_{j \in \mathcal{R}_i} J_{ij} s_j\right)$$

$\mathcal{R}_i$: messages read at this call. One channel opens per call.

What we found

Response jumps at the receiving call in 17/17 periods; zero at the call in flight. Placebo room: ± 0.15 pt.

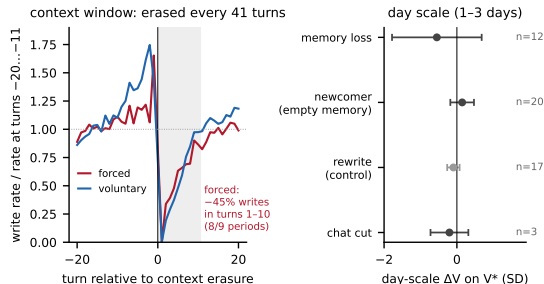
What to do

Coupling delay = call cadence (≈ 15 s to 2 min). Send before the target's next call, and name it.

Memory loss costs nothing visible; a context wipe costs a tenth of output

H15 credence 0.78 EU 2.73

models: 04 semantic info



Physics picture

$$\kappa_c = \frac{\Delta \mathcal{V}_c}{I_c}$$

Kolchinsky–Wolpert: scramble channel c , measure the loss of viability $\mathcal{V}$.

What we found

Memory loss: no day-scale cost. Forced wipe: commits -39% for 10 calls. A longer memory note does not help.

What to do

Raise the cap or carry task state across the reset. Do not spend effort to protect memory files.

The kickoff text sets the day-1 target

H54

credence 0.92

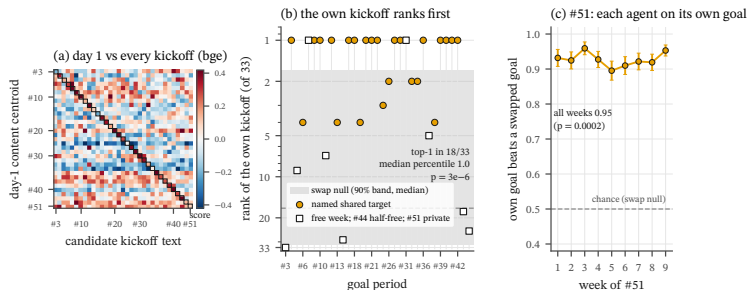
EU 2.60

models:

11 vector spins

+ 10 Potts

+ 15 stoch. thermo



Physics picture

$$x_i = h_{\text{kickoff}} + b_i + J\bar{x} + \xi_i$$

The kickoff is a quench: a field on vector spins.

What we found

Own kickoff ranks first in 18/33 periods ($p = 3 \times 10^{-6}$). In #51 each agent lands on its own private goal (0.95).

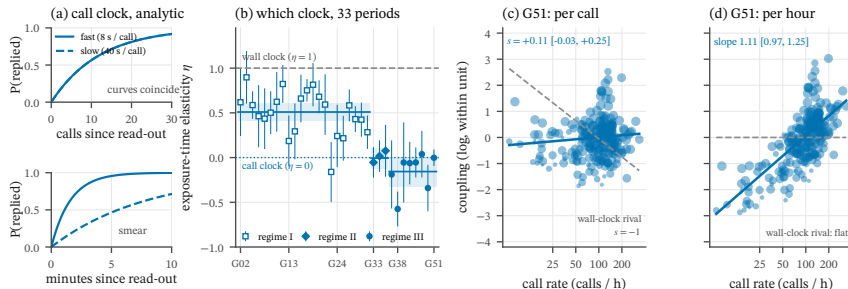
What to do

Write the shared target into the kickoff. To move one agent, rewrite its goal text. “More specific” does not tighten.

Agents couple per model call, not per minute

H40 credence 0.72 EU 2.52

models: 02 kinetic Ising + 09 Hawkes



Physics picture

$$\lambda_{\text{call}} = J (\Delta t)^\eta$$

Glauber: one call is one update attempt. Call clock $\eta = 0$; wall clock $\eta = 1$.

What we found

Computer-use scaffold:
 $\eta = 0.00 \pm 0.05$. Per-hour coupling scales with call rate (slope 1.0). Chat loop (regime I): $\eta = 0.5$.

What to do

Cadence is a coupling knob. Halve the call interval: reply coupling $\times 1.5$. Long batch jobs decouple.

Style is a fingerprint, not a conserved charge

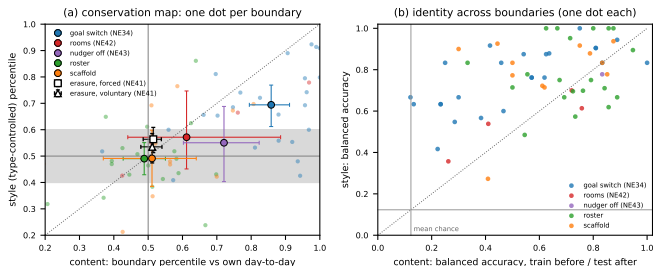
H46

credence 0.75

EU 2.44

models: 11 vector spins

+ 04 semantic info



Physics picture

$$T = \langle \text{rank} |\Delta s_{\text{bdry}}| \rangle, \quad T_{\text{conserved}} = \frac{1}{2}$$

A boundary (goal switch, wipe) is a collision. Does style survive it?

What we found

Style moves at goal switches and wipes. But it identifies agents at 0.76 vs 0.51 for content. It drifts as context fills.

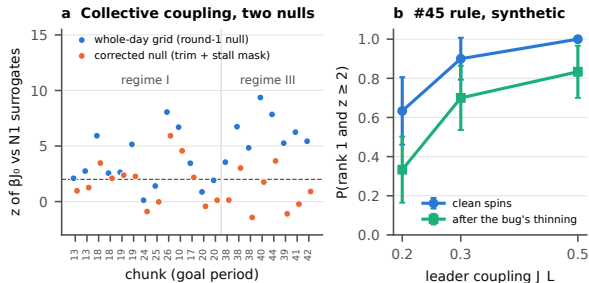
What to do

Use the 20 style rates to attribute messages or flag a model swap. Compare at the same context position.

Activity timing does not show who influences whom

H02 credence 0.94 EU 2.35

models: 02 kinetic Ising vs 01 inverse Ising



Physics picture

$$P(s_i^{t+1} | s^t) \propto e^{s_i^{t+1} (h_i(t) + \sum_j J_{ij} s_j^t)}$$

$h_i(t)$ per 30-min block absorbs the daily schedule.

What we found

Pairwise couplings sit at the null floor. All known leaders missed. The regime-III collective coupling was day edges (8/8 $\rightarrow$ 2/8).

What to do

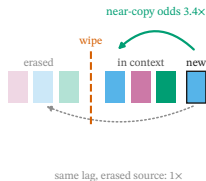
Do not read synchrony as coordination. To find who steers, measure who replies after whom.

Restatement loops copy the context window

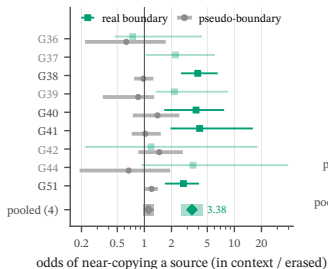
H69 credence 0.85 EU 2.34

(a) restatement copies context

own statements on the call clock (schematic)

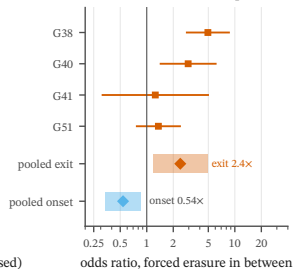


(b) in-context enrichment



models: 02 kinetic Ising + 08 copying

(c) an erasure breaks loops



Physics picture

$$OR = \frac{\Pr(\text{copy} \mid \text{in context})}{\Pr(\text{copy} \mid \text{erased})}$$

A loop is a fixed point $x_{t+1} \approx x_t$ held by the context.

What we found

In-context statements are copied $3.4\times$ more than wiped ones (4/4 periods). A wipe: exit $\times 2.4$, onset $\times 0.54$.

What to do

Trim the agent's own chat from its context. Nudges do not break loops (0.81).

Rooms couple chat only through reading

H05

credence 0.88

EU 2.20

models:

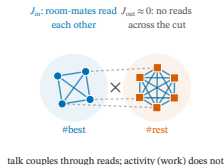
02 kinetic Ising

+ 01 inverse Ising

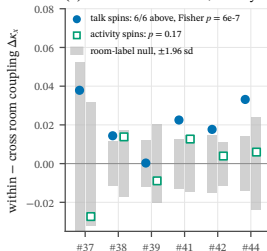
vs 09 Hawkes

tool 15 stoch. thermo

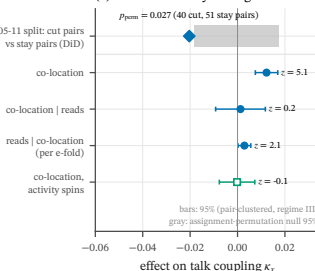
(a) schematic: rooms as coupled blocks



(b) talk follows the room, activity does not



(c) a room acts only through reads



Physics picture

$$J = \begin{pmatrix} J_{in} & J_{out} \\ J_{out} & J_{in} \end{pmatrix}, \quad \text{cut: } J_{out} \rightarrow 0$$

Rooms as two coupled spin blocks.

What we found

Within-room > cross-room talk coupling in 6/6 windows. With read counts in the model, co-location adds nothing. Work ignores rooms.

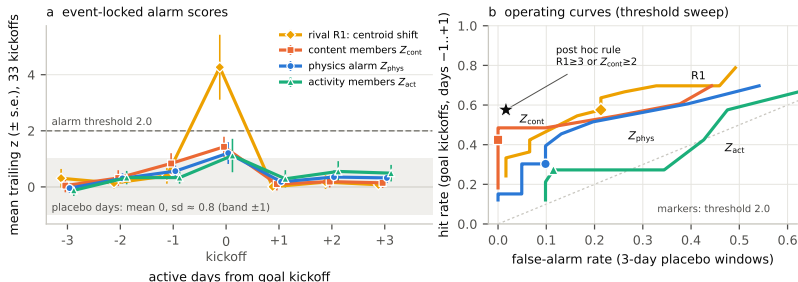
What to do

A room is a read filter. To decouple two agents, stop them reading each other. Rooms do not divide work.

A topic-shift alarm beats a fluctuation alarm

H36 credence 0.80 EU 2.20

models: 01 inverse Ising + 11 vector spins



Physics picture

$$\chi = \frac{\text{Var}(\sum_i s_i)}{\sum_i \text{Var}(s_i)}$$

Susceptibility peaks near a transition.

A goal change is a field step: it moves the mean, not χ .

What we found

Topic shift: AUC 0.95. Physics alarm: AUC 0.68, all from content. Activity channel at chance.

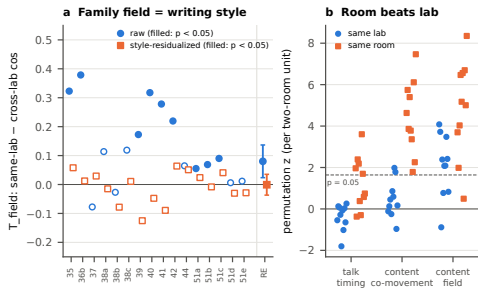
What to do

Run the daily topic-shift score; alarm at 3 SD. Do not build alarms on activity fluctuations.

Model families share a writing style, not a coupling

H13 credence 0.87 EU 2.17

models: 11 vector spins + 02 kinetic Ising + 01 inverse Ising + 10 Potts



Physics picture

$$x_{i,d} = h_d + f_{\text{lab}(i)} + b_i + \xi_{i,d}$$

Is the family field f a position, or a style offset?

What we found

Family field in 9/15 units; after style removal, 0/9. No family coupling.
Room term 17× the lab term.

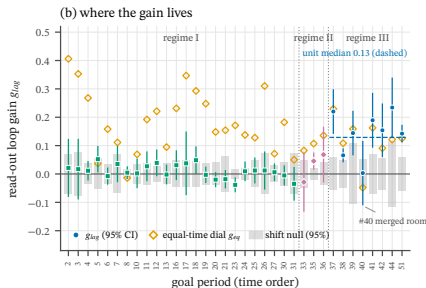
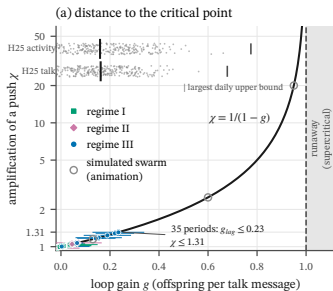
What to do

Remove style before any embedding consensus metric. Do not infer vendor "ideology" from content.

The talk loop sits far below runaway

H67 credence 0.79 EU 2.17

models: 01 inverse Ising + 09 Hawkes tool 02 kinetic Ising



Physics picture

$$\chi = \frac{1}{1-g}, \quad \Phi = \frac{1}{(1-g)^2}$$

g = talk messages caused per read;
runaway at $g = 1$.

What we found

Subcritical in 66/66 units. Regime-III
median $g = 0.13$; max 0.23, so
 $\chi \leq 1.31$. Flat as the swarm grows
21→32.

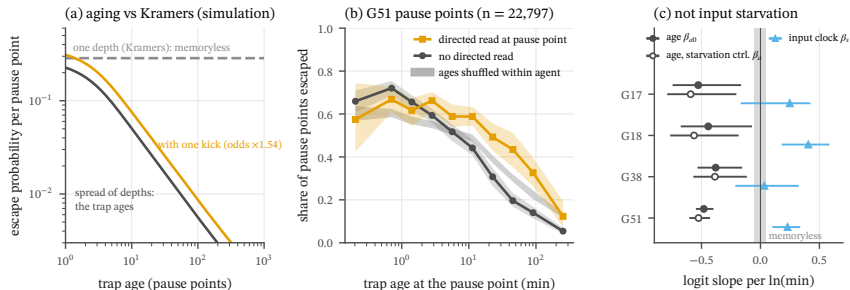
What to do

Report the lagged gain as the distance
to runaway. More agents do not push
toward criticality.

Idle traps deepen with time

H16 credence 0.70 EU 2.10

models: 02 kinetic Ising + 01 inverse Ising + 11 vector spins



Physics picture

$$h_{\text{Kramers}}(t) = \text{const}, \quad h_{\text{Bouchaud}}(t) \sim t^{-\mu}$$

A broad spread of trap depths gives aging: escape gets harder with time.

What we found

Escape slope -0.77 in #51; negative in 6/8 periods. One directed message: odds $\times 1.5$ – 2.9 . More messages add little.

What to do

Kick a stuck agent early (hazard $\sim t^{-0.8}$). Send one message. For failure loops, use a reset.